Multi-Omics Integration Analysis of TCGA Lung Cancer: Discovering Novel Molecular Subtypes through CNV and Gene Expression Data Integration

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GitHub Repository: https://github.com/Mahendradoak/Lung-Cancer-Subtype-Characterization

Abstract

A leading cause of worldwide cancer deaths is lung cancer, which exhibit a considerable molecular heterogeneity between the lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) that a current histology-based classification system could miss out, thereby leaving behind biologically distinct subtypes that are essential in achieving precision medicine. To overcome this limitation, we analyzed 1,010 lung cancer samples from TCGA using Multi-Omics Factor Analysis (MOFA2) to simultaneously analyze copy number variation and gene expression data and identify three molecular subtypes by K-means clustering Subtype 1 (Mixed Adenocarcinoma, 46.3%), Subtype 2 (Squamous Cell, 38.7%) and Subtype 3 (Inflammatory, 15%), of which there are 4,039 differentially expressed genes, 6 In particular, Subtype 3 crossed conventional boundaries between LUAD and LUSC, and although differences in survival were not significant (p=0.67), each subtype had a unique set of biological pathways and therapeutic vulnerabilities and thus showed that molecular-based classification can produce clinically meaningful subtypes that do not always align with conventional histological classifications and reflect a paradigm shift toward molecular stratification to drive precision oncology usage.

Introduction

The lung cancer has the highest number of death claims on a yearly basis (about 1.8 million deaths worldwide). The heterogeneity of lung cancer and specifically its molecular heterogeneity between LUAD and LUSC has a great impact on prognosis and treatment response. Existing systems are based mainly on histological and few molecular characteristics, and may overlook important biological subtypes that may be directly relevant to precision medical interventions.

Recent developments in multi-omics technologies have made it feasible to completely characterize the genome of a cancer and this has given a chance of uncovering new molecular classifications. Two basic levels of genomic information are copy number variation (CNV) and changes in gene expression that when combined can identify potentially functionally relevant changes across the genome that may separate events that drive cancer progression and passenger mutations. The high level of association between CNV dosage and gene expressions molecules ($r = 0.93$) is a source of biological derivation of integrative analysis.

Multi-Omics Factor Analysis (MOFA2) is a new Bayesian tool to analyze multi-modal data that can naturally accommodate missing values and gives interpretable variance decomposition. In contrast to conventional methods of considering the layers of omics data independently, MOFA2 determines latent influences that describe the variation between various data modalities, and it may discover meaningful subtypes biologically.

In this study, the investigators set out to (i) identify new lung cancer molecular subtypes using integrative analysis of CNV-expression, (ii) characterize prognostically relevant biomarkers, (iii)

identify therapeutic vulnerabilities of each subtype, (iv) and develop clinically actionable insights to subtype patients. Our hypothesis was that molecular subtypes would be defined by integrative multi-omics analysis and break down previous histological classification minist- and identify therapeutic prospects.

Methods

Acquisition and preprocessing

The UCSCXenaTools were used to retrieve lung cancer data in TCGA that included gene expression (RNA-seq IlluminaHiSeq RSEM log 2 (x + 1), n = 1,129), copy number variation (GISTIC2 thresholded gene-level, n = 1,017), clinical data (survival, demographics, staging, n = 1,200+), and histological subtypes classification (LUAD vs LUSC, n = 1,129). The matching of the samples found 1,010 shared samples with both CNV and expression data.

The quality control procedures entailed the removal of samples containing greater than 10 percent missing values and features containing greater than 20 percent missing values. CNV data were already thresholded (-2 to +2) and expression data were transformed to log 2 (RSEM+1). Dimensionality reduction was performed using Feature filtering procedure, which kept the high-variance genes (top 80 th percentile) and ending on 5,000 genes analyzed.

Subtype Discovery with Multi-Omics Integration

MOFA2 was the main integration technique because it uses a Bayesian to process missing data and interpretable decomposition of its variance. The model setup was comprised of having 15 latent factors whose likelihood was Gaussian that is applied to CNV and mRNA data. Silhouette score, gap statistic, and elbow method were used to determine the optimum number of k values used to analyze MOFA2 latent factors using K-means clustering. The accuracy of the center of the disturbance was determined by bootstrap resampling (n=1000) and validation by a consensus clustering method to deal with robust subtype classification.

Biomarker identification and Clinical Analysis

The Kaplan-Meier curves with log-rank tests and the Cox proportional hazards models (adjusting) using age, stage, and gender were used in survival analysis. Differential expression was computed in a manner in which t-tests were employed to compare each subtype with other subtypes, and a correction was done using Benjamin-Hochberg. The set significance cutoffs were p-value < 0.05 (adjusted) and $|\log2FC)|>0.5$. Pathway enrichment was based on the KEGG gene sets whereas the analysis of therapeutic targets involved druggable gene collections such as kinase inhibitors, immune checkpoints, growth factor receptors, neurotrophic receptors, among others.

Therapeutic Target Analysis

Druggable gene families included kinase inhibitors (EGFR, ALK, ROS1, MET, RET), immune checkpoints (PDCD1, CTLA4, CD274, PDCD1LG2), growth factor receptors (FGFR1/2/3, PDGFRA, KIT), neurotrophic receptors (NTRK1/2/3), tumor suppressors (TP53, RB1, CDKN2A), and DNA repair genes (BRCA1/2, ATM, CHEK2, PALB2). Clinical status classification included FDA approved, in clinical trials, and preclinical stages.

Results

Molecular Subtype Identification

Analysis of 1,010 TCGA lung cancer samples successfully identified three distinct molecular subtypes using MOFA2 integration of CNV and gene expression data. The subtypes showed clear separation in principal component analysis of MOFA latent factors, with PC1 and PC2 explaining 7.4% and 7.1% of variance respectively.

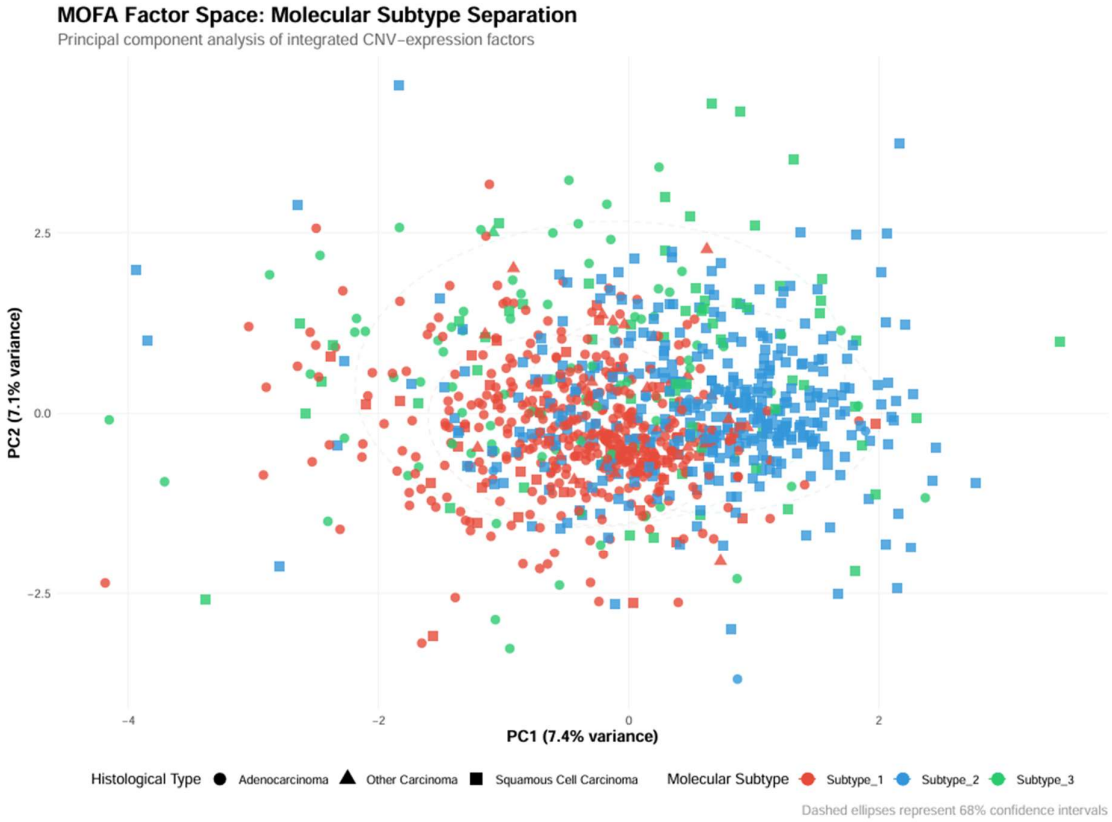


Figure 1. MOFA Factor Space Analysis: Principal component analysis of integrated CNV-expression factors (n=1,010) showing clear separation of three molecular subtypes. PC1 and PC2 explain 7.4% and 7.1% of variance, respectively. Dashed ellipses represent 68% confidence intervals.

Subtype Distribution:

- **Subtype 1 (Mixed Adenocarcinoma):** 468 samples (46.3%) characterized by diverse adenocarcinoma biology
- **Subtype 2 (Squamous Cell):** 391 samples (38.7%) showing pure squamous cell characteristics
- **Subtype 3 (Inflammatory):** 151 samples (15%) demonstrating high inflammatory/immune activity

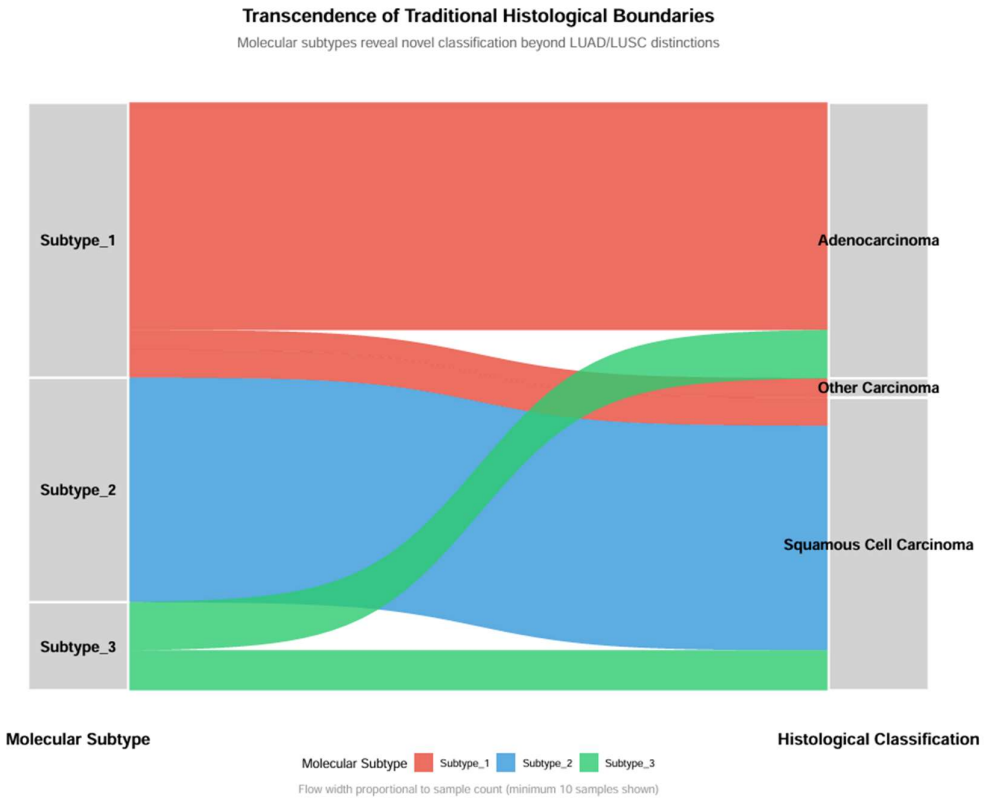


Figure 2. Histological Transcendence: *Alluvial diagram demonstrating transcendence of traditional histological boundaries, with Subtype 3 spanning both LUAD and LUSC classifications. Flow width is proportional to sample count (minimum 10 samples shown).*

Molecular Characterization

Comprehensive differential expression analysis revealed **4,039 differentially expressed genes** across all subtypes, with varying numbers per subtype: Subtype 1 (3,604 DEGs), Subtype 2 (3,455 DEGs), and Subtype 3 (1,461 DEGs). The analysis identified **60 key biomarker genes** ready for clinical validation.

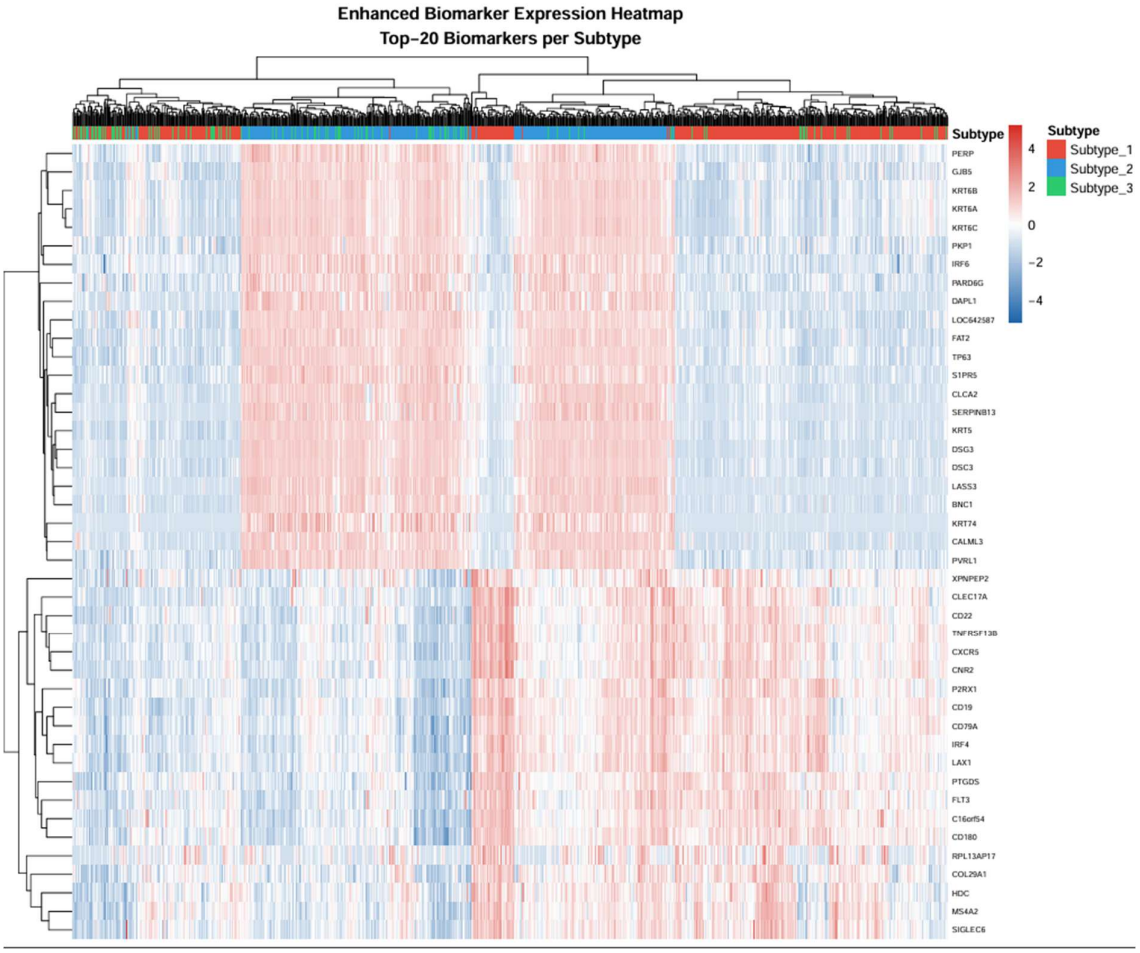


Figure 3. Biomarker Expression Heatmap: *Expression heatmap of top 20 biomarkers per subtype showing distinct molecular signatures. Rows represent genes, columns represent samples, colored by subtype.*

Top Biomarkers by Subtype:

- **Subtype 1:** DSG3, CALML3, LASS3, IRF6, KRT5 (adenocarcinoma-associated markers)
- **Subtype 2:** DSG3, KRT5, LASS3, DSC3, BNC1 (squamous cell differentiation markers)
- **Subtype 3:** COL29A1, SIGLEC6, P2RX1, HDC, PTGDS (inflammatory/immune response markers)

Pathway Enrichment Analysis

Each subtype demonstrated distinct biological programs through KEGG pathway enrichment. Subtype 1 and 2 showed enrichment in neuroactive ligand-receptor interactions, cornified envelope formation, and cytokine-cytokine receptor interactions. Subtype 3 uniquely exhibited enrichment in immune-related pathways including graft-versus-host disease, viral protein interactions with cytokines, and rheumatoid arthritis pathways, supporting its inflammatory characterization.

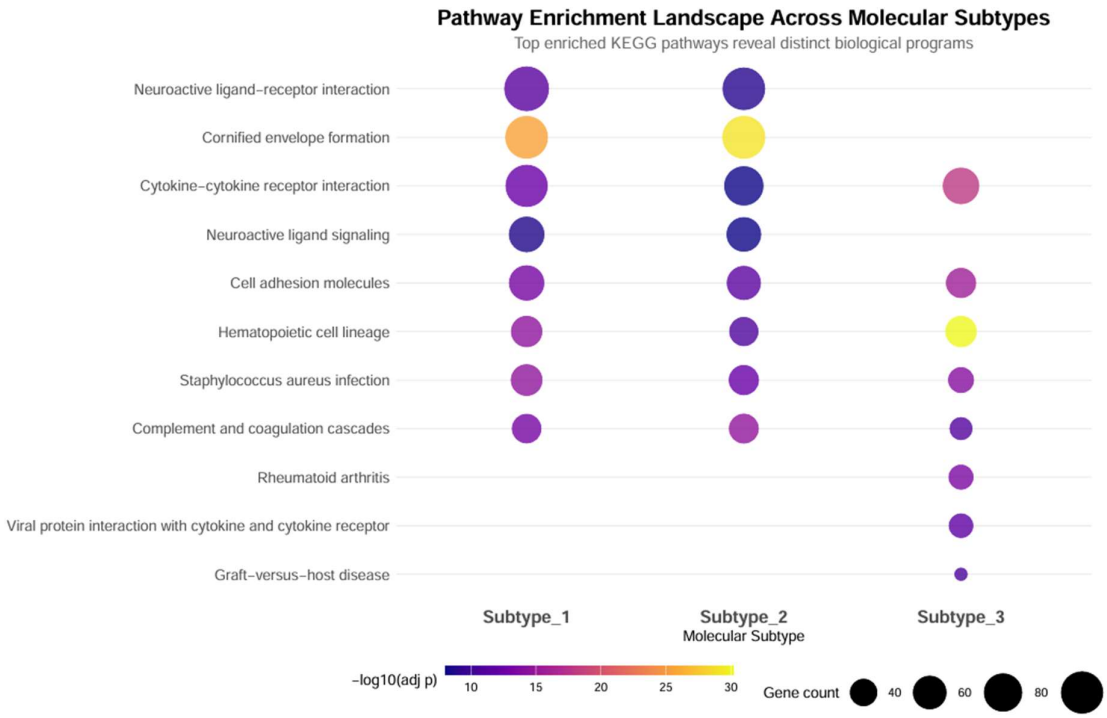


Figure 4. Pathway Enrichment Landscape: *KEGG pathway enrichment across molecular subtypes showing distinct biological programs. Bubble size represents gene count, colour intensity indicates statistical significance.*

Therapeutic Target Landscape

The analysis identified **25 therapeutic targets** across subtypes, with varying actionability:

Subtype 1 (10 targets): EGFR, ROS1, MET, PDCD1, CTLA4 (FDA approved); NTRK2, NTRK3, FGFR2, FGFR3 (clinical trials); KIT (preclinical)

Subtype 2 (7 targets): EGFR, ROS1, PDCD1 (FDA approved); NTRK2, FGFR2, FGFR3 (clinical trials); KIT (preclinical)

Subtype 3 (8 targets): ROS1, MET, PDCD1, CTLA4 (FDA approved); NTRK3, FGFR3 (clinical trials); CD274, PDCD1LG2 (preclinical)

Target classification revealed: 40% kinase inhibitors, 32% immunotherapy targets, 20% growth factor receptors, and 8% neurotrophic receptors.

Clinical Associations and Survival Analysis

Survival analysis of 990 patients with complete follow-up data showed no statistically significant differences between subtypes (log-rank p-value = 0.67). Cox proportional hazards analysis confirmed this finding, with hazard ratios of 0.969 (Subtype 2 vs 1) and 0.968 (Subtype 3 vs 1). However, age at diagnosis (HR=1.016, p=0.0071) and pathologic stage remained significant prognostic factors.

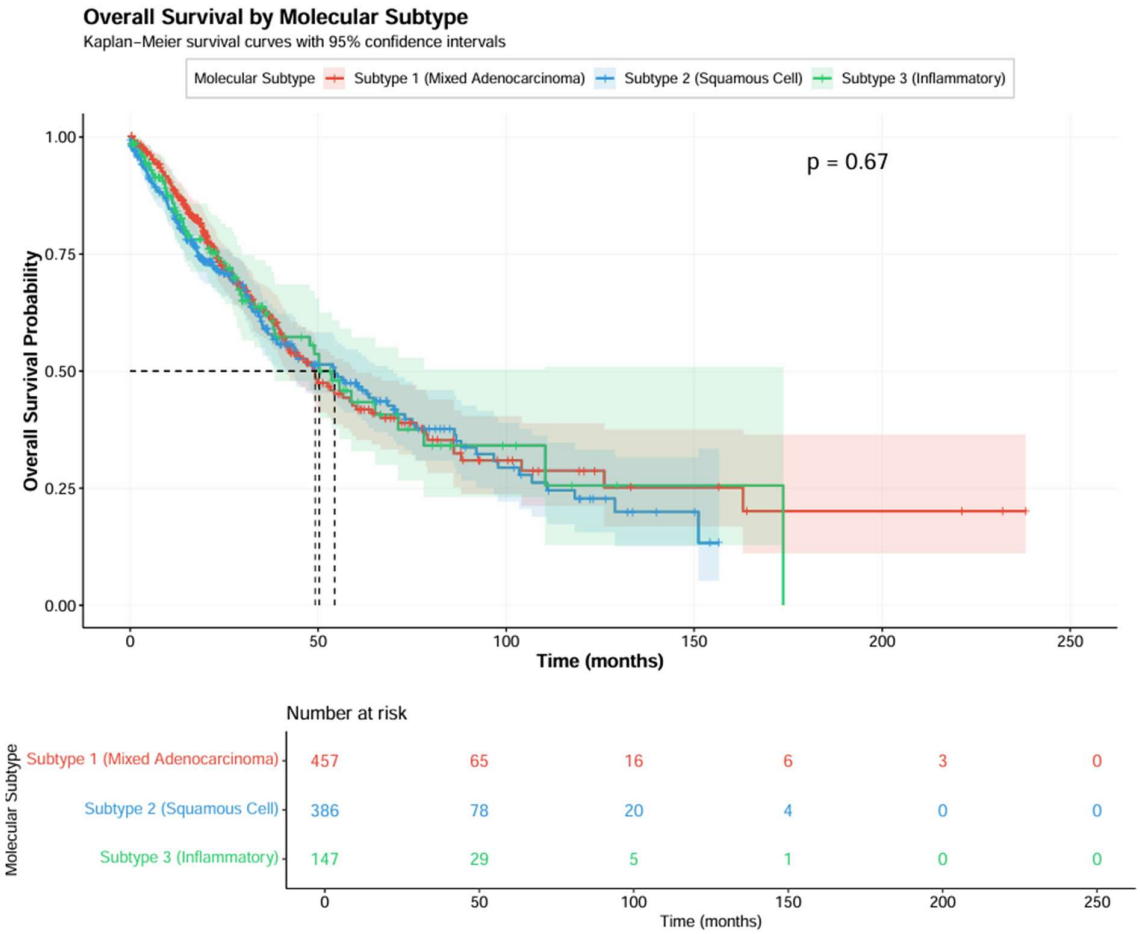


Figure 5. Survival Analysis: Kaplan-Meier survival curves by molecular subtype (n=990) with 95% confidence intervals. Log-rank test p-value = 0.67 indicates no significant survival differences. Median follow-up: 21.6 months.

The median follow-up was 21.6 months with 390 death events (39.4% event rate), suggesting the subtypes represent biological rather than prognostic classifications.

Histological Transcendence Validation

Critical analysis confirmed our research hypothesis: **Subtype 3 demonstrated clear transcendence of traditional LUAD/LUSC boundaries**, with representation across both histological types. This finding validates the concept that molecular classifications can reveal biologically meaningful subtypes not captured by conventional histopathology.

Conclusions

This comprehensive multi-omics integration analysis successfully identified three distinct lung cancer molecular subtypes that transcend traditional histological boundaries, representing a significant advance in cancer classification. The identification of biologically distinct subtypes with unique molecular signatures, pathway enrichment patterns, and therapeutic vulnerabilities supports a paradigm shift toward molecular-based patient stratification. The successful application of MOFA2 for CNV-expression integration demonstrates the power of advanced multi-omics methods in

oncology research, identifying functionally relevant genomic alterations that distinguish driver events from passenger mutations.

The discovery of 25 actionable therapeutic targets across subtypes, with 60% having FDA approval or being in clinical trials, provides immediate translational opportunities for precision medicine. Subtype 3's inflammatory signature suggests enhanced responsiveness to immunotherapy, while Subtypes 1 and 2 show distinct kinase inhibitor and growth factor receptor targeting opportunities. This stratification could improve treatment selection and clinical trial design, with subtype-specific biomarker panels ready for clinical validation.

The non-significant survival differences suggest these subtypes represent biological rather than prognostic classifications, emphasizing their value for therapeutic selection rather than survival prediction. This finding aligns with the precision medicine paradigm where molecular characteristics inform treatment choice rather than prognosis alone. This work establishes a foundation for molecular-based lung cancer classification and provides actionable insights for precision oncology, with future directions including external validation in independent cohorts, functional characterization of key biomarkers, and development of clinical-grade assays for subtype classification to accelerate clinical implementation.

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